Name: ______________________

PROPORTION, (PROPORTION IN CONTEXT)

Date: ________________

LO: I can solve real-world proportion problems involving recipes, maps, scales, and exchange rates.

Proportion in real life

Proportion in context means scaling quantities up or down while keeping the same ratio between ingredients, distances, or values.

Applications: Recipes (scaling ingredients), maps (scale factors), currency exchange, and mixing ratios.

Quick examples

	Scale 1:50000 means 1 cm = 50000 cm = 500 m	map scale
	Recipe for 4 for 6: multiply by 1.5	scale factor
	£1 = .25 means £40 = 0	exchange rate
	Scale 1:50000 means 1 cm = 50 m	forgot a zero: 500 m
	Recipe for 4 to 6: add 2 to	must multiply by

Key idea

Find the scale factor (new ÷ original) and multiply all quantities by it.

$$\textbf{Scale factor} = {}^{\text{new amount}}/_{\text{original}}$$

Then multiply each ingredient/measurement by SF

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – scaling a recipe

A recipe for 4 uses 200g flour, 100g sugar, 2 eggs. Make it for 6.

Scale factor: 6 ÷ 4 = 1.5

Flour: 200 × 1.5 = 300g, Sugar: 100 × 1.5 = 150g

Eggs: 2 × 1.5 = 3

300g flour, 150g sugar, 3 eggs

Multiply every ingredient by the same scale factor.

Example 2 – map scale

A map has scale 1:25000. Two towns are 8 cm apart on the map. Find the real distance in km.

8 cm × 25000 = 200000 cm

200000 cm = 2000 m

2000 m = 2 km

= 2 km

Convert from cm to km: divide by 100000.

Example 3 – currency exchange

£1 = 1.15 euros. Convert £85 to euros.

Multiply by the exchange rate

85 × 1.15

= 97.75 euros

= 97.75 euros

To convert from pounds to euros, multiply by the rate.

Example 4 – best buy using proportion

Shop A: 500ml for £2.40. Shop B: 750ml for £3.30. Which is better value?

Shop A: 2.40 ÷ 500 = 0.48p per ml

Shop B: 3.30 ÷ 750 = 0.44p per ml

Shop B is cheaper per ml

Shop B is better value

Compare unit prices to find the best deal.

YOUR TURN – PRACTICE (deliberate practice)

1. Scaling recipes
Scale the recipe from 4 servings to the target.

a) 200g flour for 4. Amount for 10?

b) 150ml milk for 4. Amount for 6?

c) 3 eggs for 4. Eggs for 12?

d) 80g butter for 4. Amount for 3?

2. Map scales
Calculate real distances from map measurements.

a) Scale 1:50000. Map distance 6 cm. Real distance?

b) Scale 1:25000. Map distance 12 cm. Real distance?

c) Scale 1:100000. Real distance 8 km. Map distance?

d) Scale 1:20000. Map distance 4.5 cm. Real distance?

3. Currency exchange
Convert using the given exchange rate.

a) £1 = .25. Convert £60 to dollars

b) £1 = 1.15 euros. Convert £200 to euros

c) £1 = .25. Convert 00 to pounds

d) £1 = 1.15 euros. Convert 460 euros to pounds

4. Mixing and ratios
Use proportion to solve.

a) Concrete mix: 1 cement : 3 sand. Need 20 kg total. How much cement?

b) Paint: 3 red : 2 white = pink. Need 2.5 L total. How much red?

c) Squash: 1 part cordial to 4 parts water. Make 2 litres. Cordial needed?

d) Model scale 1:72. Real plane 36 m. Model length?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Scale 1:25000 means 1 cm represents 250 m ☐

Correct answer: __________

Reason: _________________

b) To scale a recipe from 4 to 6, add 2 to each ingredient ☐

Correct answer: __________

Reason: _________________

c) To convert £ to euros, multiply by the exchange rate ☐

Correct answer: __________

Reason: _________________

d) Scale 1:50000 means 1 cm = 50 m ☐

Correct answer: __________

Reason: _________________

e) The scale factor for 4 servings to 6 is 1.5 ☐

Correct answer: __________

Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A recipe for **6 people** uses: 450g chicken, 200g rice, 150ml sauce, 3 peppers. You have **750g chicken** and want to use it all. How much of each other ingredient do you need? How many people does this serve?

Expression: ____________________________

Simplified: ____________________________

b) On a map with scale **1:50000**, a lake has area **12 cm²** on the map. Find the real area in km². If a runner goes around the lake and it measures **14 cm** on the map, how far do they actually run?

Expression: ____________________________

Simplified: ____________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!